

Eunsu Baek

Ph.D Student · Adaptive Sensing for DNNs / Reliable AI / Embodied AI

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Summary

Ph.D. student at **SNU GSDS** whose research focuses on building **reliable real-world AI** and **adaptive sensing for DNNs**—improving robustness by optimizing **sensor configurations** in **data acquisition**, beyond scaling models alone. This idea is inspired by how humans adapt perception via active sensing (e.g., corrective lenses or pupil dilation/constriction) in the wild. **To validate this concept**, my research spans robustness benchmarking, adaptive sensing frameworks, and a path toward next-generation AI, with first-author publications at **CVPR**, **ICLR**, and **NeurIPS**.

- **Benchmarks:** Built **ES-Studio**, a physically controllable testbed that manipulates environment and sensor factors to induce covariate shifts, and collected the **ImageNet-ES** series (394K real-camera images) to evaluate real-world robustness, demonstrating efficacy of **sensor control**.
- **Frameworks:** Developed **Lens**, an **adaptive sensing framework** that yields up to **50%** accuracy gains and enables **small models to match large-model performance (up to 50× larger) without retraining** under real-world shifts.
- **Embodied AI:** Proposed a **closed-loop adaptive sensing** framework that elevates sensing to a **first-class** component toward **robust, sustainable, and equitable AI**, integrating **active sensing agents** into perception–action loops to address sparse rewards under dynamic shifts.

I have also contributed to collaborative projects published at **UbiComp**, **ESWA**, **WACV**, and **MobiSys**, released research artifacts as **open-source contributions** (e.g., OpenOOD), and received awards in competitions and top-tier venues. With experience in full-stack **AI system prototyping**, **dataset engineering**, and **wearable AI applications**, I have **mentored 7+ students** across sensing, vision, and interaction. My long-term goal is to build **trustworthy embodied AI systems** that learn to perceive and act adaptively in the real world toward scalable, human-aligned intelligence.

Education

Seoul National University

Ph.D. in Data Science (Advisor: Dr. Hyung-Sin Kim)

- Research Topic: Adaptive sensing for DNNs, Reliable AI, Embodied AI

Seoul, Republic of Korea

Mar. 2022 – Present

Seoul National University

M.S in Computer Science and Engineering (Advisor: Dr. Jinwook Seo)

- Research Topic: Human-Computer Interaction, Explainable AI (XAI)

Seoul, Republic of Korea

Mar. 2020 – Feb. 2022

Sookmyung Women's University

B.S. in Mathematics (double major in Computer Science)

Seoul, Republic of Korea

Mar. 2013 – Feb. 2018

Publications

Conferences & Journals (* Co-first authors, † Corresponding authors)

- [C.4] *Simulating Environment and Sensor Domain Perturbations via Generative Image Translation*
Jiyeon Kim, Eunsu Baek, Hyung-Sin Kim[†]
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2026, Tucson AZ, USA.
- [C.3] *AI Should Sense Better, Not Just Scale Bigger: Adaptive Sensing as a Paradigm Shift*
Eunsu Baek, Keondo Park, Jeonggil Ko, Min-hwan Oh, Taesik Gong, Hyung-Sin Kim[†]
Conference on Neural Information Processing Systems (NeurIPS) 2025, San Diego CA, USA. (**Position**, Acceptance ratio: **8%**) [pdf] [poster]
- [J.2] *ColonOOD: A Complete Pipeline for Optical Diagnosis of Colorectal Polyps Integrating Out-of-Distribution Detection and Uncertainty Quantification*
Sehyun Park^{*}, Dongheon Lee^{*}, Ji Young Lee, Jaeyoung Chun, Ji Young Chang, Eunsu Baek, Eun Hyo Jin, and Hyung-Sin Kim[†]
Expert Systems with Applications (ESWA), Vol. 295, Jan. 2026 (IF **Top 7%** in Operations Research) [pdf]
- [C.2] *Adaptive Camera Sensor for Vision Model [Lens]*
Eunsu Baek, Taesik Gong[†], Hyung-sin Kim[†]
International Conference on Learning Representations (ICLR) 2025, Singapore [pdf] [poster], [github]
- [C.1] *Unexplored Faces of Robustness and Out-of-Distribution: Covariate Shifts in Environment and Sensor Domain [ImageNet-ES]*
Eunsu Baek, Keondo Park, Jiyeon Kim, Hyung-sin Kim[†]
IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR) 2024, Seattle WA, USA [pdf] [poster] [video] [github]
- [J.1] *MIRROR: Towards Generalizable On-Device Video Virtual Try-On for Mobile Shopping*
Dong-sik Kang, Eunsu Baek, Sungwook Son, Youngki Lee, Taesik Gong, Hyung-sin Kim[†]
PACM IMWUT, Vol. 7, No. 4, Dec 2023; presented at UbiComp 2024, Melbourne [pdf] [video1] [video2]

Demos

[D.1] *On-Device Video Virtual Try-On for Mobile Shopping* (People's Choice Demo Runner-up Award)

Dongha Ahn, Dong-sik Kang, Eunsu Baek, Hyung-sin Kim[†]

ACM International Conference on Mobile Systems, Applications, and Services (Mobisys) 2024, Tokyo, Japan

[pdf] [video]

Ongoing Manuscripts

[O.2] "Annoymized"

Eunsu Baek, Dayeon Woo, Jinmo Yang, Hong-Jun Kim, Rian Kim, Woojin Jang, Taesik Gong, Hyung-Sin Kim[†]

Under review at CVPR 2026

[O.1] "Annoymized"

Eunsu Baek, Jiyeon Kim, Keondo Park, Sojung Kim, Taesik Gong, Hyung-Sin Kim[†]

Under review at CVPR 2026

Research Experience

AIoT Lab at SNU (Advisor: Hyung-sin Kim)

Ph.D. student

Seoul, Republic of Korea

Jul. 2022 – Present

- **Team leader of Adaptive sensing for DNNs** — Improved robustness to covariate shifts by *optimizing sensor parameters* in a novel paradigm.
- **Lead author & developer of ImageNet-ES (Luminous & Diverse)** — Built a controllable testbed (*ES-Studio*), collected 202K/192K covariate-shifted images in environment & sensor domains; revealed limits of training-based robustness; validated camera sensing as an effective remedy.
- **Lead author & developer of Lens** — *The first sensing framework* adapting to model and scene for robust prediction under real-world noise.
- **Lead author of NeurIPS Position** — *Advocated a paradigm shift* in AI from scaling models to *adaptive sensing*, toward sustainable, robust, and equitable intelligence; conducted in *collaboration with four domain-expert* across AI, systems, reinforcement learning, and edge intelligence.
- **Co-author MIRROR** — Designed system & user scenarios for a *generalizable on-device VTO*; contributed to manuscript writing and visuals.
- **Co-author of ColonOOD** — Developed CAD pipeline and co-wrote manuscript on a *polyp diagnosis system* with OOD and uncertainty modeling.
- **Co-author ImageNet-sES** — Defined simulation strategies for *ImageNet-ES*; mentored a master's student on generative modeling and analysis.
- **Co-authored proposals & served as a core researcher in national projects** — *Surveillance AI, Embodied AI, XR interaction* (NRF/IITP).
- **Mentored 7 students** in adaptive sensing, robustness benchmarking, generative models, virtual try-on (VTO) applications, and industrial vision.
- **Committee member of GSDS VLM dataset challenge** — Contributing to dataset creation and competition logistics.

HCI Lab at SNU (Advisor: Jinwook Seo)

M.S. student & Research Intern

Seoul, Republic of Korea

Oct. 2019 – Feb. 2022

- **Lead author & developer of MRViz** — Visual analytics system for CNN reliability; user studies & adversarial case analysis for robustness insights.
- **Lead author & developer of NutriSeeON** — Vision-based assistive app with wearables enabling blind-friendly interaction for grocery shopping.
- **Involved in IITP-funded collaborative projects** — *XAI-based neuro-signal analysis (ECG)* and *FPGA-accelerated visualization*.
- **Collaborator of Tangible Illusion** — Ran user studies validating a VR system simulating gravity.

HCS Lab at SNU (Advisor: Young-ki Lee)

Research Intern

Seoul, Republic of Korea

May. 2019 – Sep. 2019

- **Main author & developer of SilverStone** — Developed *homomorphic encryption* with layer-wise computation for offloaded DNN inference.

Honors & Awards

2024	Best Paper Award , ImageNet-ES (CVPR 2024) — \$755	SNU BK21+
	People's Choice Demo Runner-up , On-Device VTO Demo	Mobisys 2024
	Travel Grant , Conference Attendance Support — \$400	CVPR 2024
2023–2024	Medical AI Scholarship , — \$3,805 / semester	SNU AIMED
2023	1st Place (212 teams) at Samsung AI Challenge , Camera-Invariant Domain Adaptation — \$7,611	SAIT
2022	1st Place at Coral Competition , NaviBI: Navigation for Blinds [slides], [video] — \$1,522	Google & SNU
	GSDS Development Fund , Software Globalization Support — \$7,611	SNU GSDS
2017	Outstanding Award , Debate Competition: "AI Apocalypse"	SMWU
2016	Leadership Scholarship , President, College of Natural Sciences; Emergency Council (GSC vacant) — \$3,805 / semester	SMWU
2015–2018	Samsung SW Scholarship , Academic Excellence in CS — \$266 (×5)	Samsung
	Academic Excellence , 1st in 21 major courses (GPA 4.07/4.3) — \$2,891 (×3)	SMWU
2014	Club Innovation Award , Captain, Natural Science Research Society — \$380	SMWU

Academic & Open-source Contributions

2025	Reviewer, NeurIPS 2025 Workshop on Learning to Sense , Invited Program committee member [L2S]
2025 – Present	Committee of GSDS Dataset Challenge for Language Models , Organizing dataset creation and competition logistics.
2025	Invited Reviewer , Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT).
2025	ImageNet-ES Diverse Release , ImageNet-ES variant of non-luminous scenes under diverse environment settings (192K)
2024	ImageNet-ES Integration to OpenOOD Framework , Contribute to provide standardized OOD evaluation. [openOOD]
2024	ImageNet-ES Release , Sensor control & Robustness benchmark with 202K perturbed images in ES domain [gitHub]

Representative Coursework Projects

Crowd2Crowd (C2C) — Bidirectional dataset platform for custom requests with automatic QC <i>Led & built full-stack development of dataset marketplace with bidirectional match & automated quality control. [github] [slides][video]</i>	2023 SNU GSDS
NaviBl — Navigation System for Blinds (1st Place Award) <i>Led & Built AI-driven auditory guidance system for real-time crosswalk navigation on Coral board. [slides] [video]</i>	2022 Google Coral Competition
TamrAtlas — Visual Explorer of Instagram Trends in Jeju <i>Led full-stack D3/React development on 200K crawled posts; built custom UI for search & trend exploration. [github] [slides] [video1]</i>	2021 SNU CSE
NutriSeeON — Virtual eyes to shop healthy groceries <i>Led & built vision-based Android app with wearable haptic-auditory feedback for blind-friendly grocery shopping. [github] [video]</i>	2020 SNU CSE

Talks & Presentations

NeurIPS 2025 Poster Presentation <i>AI Should Sense Better, Not Just Scale Bigger: Adaptive Sensing as a Paradigm Shift</i>	Dec. 2025 San Diego, USA
ICLR 2025 Poster Presentation <i>Adaptive Camera Sensor for Vision Models</i>	Apr. 2025 Singapore
ResearchTrend Connect Invited Talk <i>Vision-Language Meetup: Sensing for DNNs</i>	Mar. 2025 Online
UbiComp 2024 Oral Presentation <i>MIRROR: Towards Generalizable On-Device Video Virtual Try-On for Mobile Shopping.</i>	Sep. 2024 Melbourne, Australia
Trustworthy AI Workshop Invited Talk <i>MIRROR: Towards Generalizable On-Device Video Virtual Try-On for Mobile Shopping.</i>	Sep. 2024 Melbourne, Australia
CVPR 2024 Poster Presentation <i>Unexplored Faces of Robustness and Out-of-Distribution: Covariate Shifts in Environment and Sensor Domains (CVPR 2024)</i>	Jun. 2024 Seattle, USA
Mobisys 2024 Demo Presentation (Runner-up Award) <i>On-Device Video Virtual Try-On for Mobile Shopping</i>	Jun. 2024 Tokyo, Japan

Teaching Experience

Coursework

Summer 2025	Head Instructor , 7-week Project Mentoring Program, Data Analysis Boot Camp	HYUNDAI MOTORS
2024–2025	Head Lab Instructor , On-device AI Program (4 semesters)	SAMSUNG
Summer 2024	Head Lab Instructor , Advanced Expert Course in Big Data, Fintech, and AI	SNU KDT
Winter 2023	Lab Instructor , Ambient AI Boot Camp [video]	SNU GSDS
Fall 2022	Head Lab Instructor , Ambient AI Platforms and Practice (M3239.005900) [video]	SNU GSDS
Fall 2021	Lab Instructor , Digital Computer Concept and Practice (035.001)	SNU CSE
Fall 2019	Lab Instructor , Computer Programming (M1522.000600)	SNU CSE

Mentored Students

2025 – Present	Jun-Won Park (Lead Manager, KIA) , E2E Vehicle Inspection Pipeline (No Unlabeled Data)
2025 – Present	Jin-Hwan Kim (Manager, KIA) , Out-of-Distribution Data Generation for Vehicle Inspection
2025 – Present	Rian Kim (Undergrad, SNU) , Adaptive Sensing for DNNs
2025 – Present	Claire Shin (Undergrad, Princeton) , Adaptive Sensing for DNNs
2023 – 2025	Jiyeon Kim (Master, SNU) , Real perturbation simulation
2024	Dong-ha Ahn (Undergrad, SNU) , Adaptive Sensing for DNNs, Virtual Try-On
2023	Mingyun Seo (Undergrad, SNU) , Adaptive Sensing for DNNs

Skills

AI / ML	Computer Vision, Edge AI, Generative Models, VLMs, XAI, Representation Learning, RL
Systems & Prototyping	Linux, Android, Jetson, Coral, BLE, FPGA, Arduino; Testbed Construction, Rapid Prototyping
Programming	Python, C/C++, JavaScript, Java, SQL, R, Mathematica
Frameworks & Tools	PyTorch, TensorFlow, Flask, React, Vue, OpenOOD, MongoDB, Docker, Git, Neo4J, BigQuery
Research & Analysis	Dataset Curation, HCI, Data Visualization (D3, Vega, Matplotlib), Model Debugging, User Studies, Signal Processing
Math Foundations	Linear Algebra, Real Analysis, Probability & Statistics, Differential Equations, Cryptography
Languages	Korean (Native), English (Fluent)